



Emergency Isolation and Hibernation Guidance

Introduction

This Guide provides ideas for you and your family during hibernation situations – whether the hibernation is self-imposed, or where it results from a government directive. The need to hibernate – or shelter in place – can result from a wide range of crisis situations, including pandemics, natural disasters or humanmade security incidents. Crisis situations can affect your ability to move, work, communicate, socialize, pray and access critical supplies. Critical services, such as power, communications (phone and internet) water, health care, fire-fighting, schooling, access to vehicle fuels, sewage and garbage removal may also be impacted at the local, state, national, and in extreme situations, up to international levels. Under some conditions a crisis may also undermine rule of law, with increases in crime, social instability and terrorism.

The guidelines offered in this document are advisory only, and must be applied against the context of your specific situation. This Guide provides ideas on how to plan for and live through a hibernation situation, focusing on critical resource identification and management. Best practices are drawn from multiple sources, and you should conduct your own research to supplement the solutions offered.

Planning for a Crisis

During crisis situations simple activities can become challenging. By developing a ‘Crisis Plan’ you and your family can more easily identify your risks and needs, and then establish your priorities. Your Plan should contain simple checklists, helping you to focus on you most likely and impactful risks, while concurrently removing a degree of uncertainty during what can be a period of stress and anxiety.

Your Plan should ask the following simple questions:

1. What are the risks?
2. When might the risks occur?
3. How will they impact me and my family?
4. How do I address these risks?
5. What support or resources do I have to meet these risks?

Fundamentally, your safety and security will revolve around:

1. Information.
2. Shelter.
3. Sustenance.
4. Health.
5. Security.

By breaking your focus into these 5 areas you can more logically prepare you and your family for the potential challenges you may face.

- Information** → You should identify useful resources which provide you accurate and timely information. This will shape the effectiveness of your decision making.
- Shelter** → You will need to ensure that you have a place where you can hibernate, potentially for a protracted period of time. It should be safe and secure.
- Sustenance** → Sustenance: You should have sufficient water and food stores to see you through a potentially protracted hibernation situation.
- Health** → You must stay healthy. This includes exercising, but also having access to medicines, medical stores and nutritional supplements. Mental health is also important.
- Security** → Where crime, social instability and terrorist risks are present then you will need to protect yourself, your shelter, and your supplies.



Creating a Risk Register and Treatment Plan

In order to identify the risks you and your family might face it can be useful to list risks out in alphabetical order. You should then grade these in terms of probability [P] and impact [I], scoring the risks between 1 and 5.

A score of 1 would be a risk which is highly unlikely to occur, or is an incident with negligible implications. A score of 5 would be a risk that will almost definitely occur, and the impact would be significant. You should then multiply both scores together, resulting in a final score of between 1 and 25.

$$P \times I = \text{Final Risk Score}$$

The higher the score, the worse the risk. Against each risk you should then list your 'Risk Control' measures. You then re-score each risk to see whether you have managed to reduce the probability of the risk occurring, and the impact if the risk does occur. A Risk Register is never static and you should review it as the situation evolves, adjusting as needed. This process helps you identify the most likely or impactful risks, and prioritize your actions accordingly.



The Crisis Communication Plan

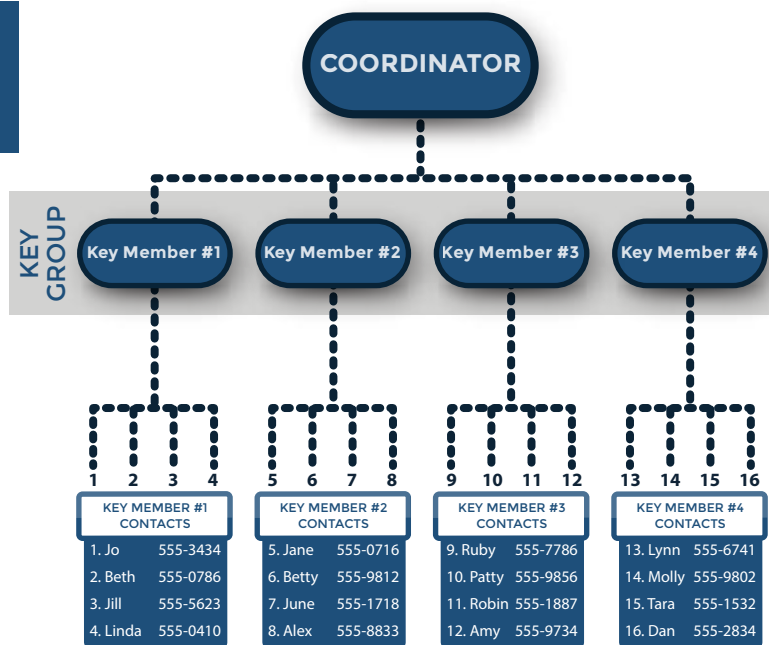
problem solving

Communication is a key aspect of everyday life, whether we enjoy talking face to face, using the telephone or the internet, or being active on social media. During a crisis being able to communicate quickly and effectively becomes more important from a safety and security perspective. We therefore need a Crisis Communications Plan in place to ensure we can communicate with the right people, quickly and effectively.

Supportive of the Crisis Communications Plan is developing a 'Phone Tree'. This is a system which allows us to quickly check in on family members, friends and other stakeholders within our community without missing anyone during a fast-moving emergency situation, and conversely without duplicating effort where someone is contacted several times. The system is simple and can be used for small family groups up to at the community level. Some key points when developing a Phone Tree system include:

- Designate 'group coordinator' who is responsible for accounting for, and reporting on, everyone within the entire group. You should always have an alternative.
- Divide your group into logical 'teams' by household, or by geographical location.
- Designate one person in that team as the 'team lead'. Within each team should be a deputy.
- During an emergency situation the team lead should confirm the status and location of everyone within their team.
- Once each team lead has accounted for those within their group, they then report to the group coordinator. This includes where people are safe, where they are injured or in harms way, or where they are missing. This then allows actions to be taken to protect people within the group who might be at risk.

Example Phone Tree Layout



Methods of Communication



While the communication infrastructure remains operational, we can use a variety of systems to communicate. Some systems allow us to create groups so that one message can be sent and read by everyone. Examples include mobile phones, laptop supported voice communications, Facebook, WhatsApp, QQ, WeChat, QZone, Tumblr, Instagram and Twitter. Other systems exist, such as:

- **HAM or Amateur Radio:** This is the probably the most commonly used system for emergency communication as it is the most powerful and versatile with a million frequencies providing a higher distance range to send and receive information. A license is required and infrastructure needs to be installed. Range distances are typically between 5 to 40 miles. With repeater systems a range of over 100 miles can be achieved.
- **CB or Citizen Band Radio:** While a useful option, this has a more limited range, with a distance normally of only 4 miles.
- **FRS Radios or Family Radio Service:** This powerful but short-ranged service has a maximum range of approximately 20 miles.
- **Scanner Radios:** These systems allow you to listen into radio transmissions in your area from different government agencies including the fire department, law enforcement, ambulance services, and other government agencies. This can provide access to reliable first responder information.
- **Multiple User Radio Service [MURS]:** This is another option with a line of sight range distance of between 3 and 20 miles, subject to local terrain and conditions.

Times should be set when check-ins will take place. This will ensure everyone carries out a welfare status notification at designated times. When agreed check-ins do not occur then this may trigger a response to ensure people are safe and secure. Where power loss is expected, or where it has occurred, then it may be necessary to preserve equipment batteries or generator use. By having pre-determined times for check-in's phones, laptops and other equipment will be able to function longer. Communications should be kept short, and equipment should then be turned off until the next check-in time.



Mapping Resources

It will be important to create a register of the resources you may need. Excel can be a useful tool to use for planning your resource needs, where you multiply the number of people against the number of days you may be in a hibernation situation. You should rationalize how supplies are broken down into meals, or the usage rates – such as how long a candle or fuel tank lasts for – for the Excel to be effective. You should also consider the rationing of resources, and ensuring that perishable items are consumed first to avoid wastage. In addition, additional assets may be required, such as freezers and propane or coal cookers. Resources should be categorised into thematic areas, such as:



- Water, milk, coffee and other drinks.
- Meats, vegetables and fruits.
- Staples items such as pasta and rice.
- Dietary supplements.
- Cooking [oils] and flavouring for food.
- Household items such as cleaning materials, toilet rolls and garbage bags.
- Lighting – including batteries and flashlights, candles, oil lamps and oils, and fire-lighters.
- Fuels and gas for vehicles, cooking equipment and emergency generators.
- Medicines and medical supplies.
- Storage containers – including freezers, boxes, jerry cans and containers.
- Purification and sanitation resources – such as masks, gloves, hand sanitizers and water purification tablets.
- Morale items – such as sweets, deserts, luxury foods, DVD movies, board games and toys for children.

Sensible stockpiling of foods that do not require refrigeration, heat, water or special preparation will extend your ability to hibernate. Canned, dried, sealed or packed foods which have an extended life-span should be bought first, and eaten last. Ready to eat meals also reduce the requirement to have cooking assets when you are sheltering in place, and are easier to move. When planning your resource needs you would also consider:

- Ensuring you have access to a manual can opener and eating utensils.
- Choosing the foods your family will eat.
- Factoring in any special dietary needs.
- Avoid foods that will make you thirsty – these include:
 - Caffeinated tea and coffee.
 - Alcohol.
 - High sodium foods like processed cheese, hot dogs, and breads.
 - High protein foods.
 - Sugary foods and drinks.
 - Disposing of garbage quickly to reduce the risk of vermin and disease.

It is important that individuals review the list of emergency supplies and consider where they live, the unique needs of their family, and any medical or special dietary needs. Should freezers break, or if power supplies fail, then frozen foods should be consumed first. And, if food resources run out, then survival will outweigh personal tastes, or societal or religious norms.



Risk of Resource Scarcity



Under crisis situations the fear of supply chain disruption can result in the general population mass buying and hoarding resources. This may overtake the ability for shops to restock in time, leading to shortages. And, where mobility restrictions are imposed, shops can quickly be depleted of their stocks. As such, early resourcing will ensure that you can

establish your emergency supplies while they are still readily available. The following may indicate the need to quickly establish your emergency stores:

- An increase in epidemic or pandemic disease transmissions.
- Elevations in the security situation, including increased social tensions, a surge in crime and terrorist attacks.
- Mobility restrictions being instigated on an international, national, State or local level.
- Military forces having leave cancelled, or mobilizing to impacted areas in response to the crisis.
- Increases in media and social media reporting on the crisis situation.
- Government alerts and warnings.
- Evidence of supplies running short within commercial outlets.
- Evidence of people evacuating from affected areas.



Freezer units will often be purchased quickly and any online orders may see delays or cancellations. Commercial outlets may impose rationing, or this may be instigated by government authorities. Fuel shortages, power disruptions and repairs to infrastructure may be impacted. Most governments will seek to maintain supply chains as long as possible – certainly in the early stages of a crisis, and this should give sufficient time for well-organized individuals to ensure they have secured the necessary hibernation supplies.

It is also important to understand during serious crisis situations a breakdown of the ‘Social Contract’ can occur, leading to increases in social instability, looting, home invasions and robberies. Under these conditions’ families should break their emergency stores into a small supply which can be given to potential aggressors, and a larger supply which is hidden from view and protected from theft.

Police and military forces deployed to ensure the safety and security of society, may also see instances where government forces are heavy handed in their response to a crisis, especially during mobility restrictions and curfews, or where they directly perpetrate criminal acts.



Staying Healthy



Most people require a minimum of 1,200 calories a day to stay healthy. People who have a strenuous fitness routine, or whose job includes manual labor, will often need more calories. Typically, men require around 2,500 calories a day and women require around 2,000 calories a day. Children's daily calorie intake requirements vary depending on their age and gender:

- Females aged 4 to 8 years typically require 1,200 calories.
- Males aged 4 to 8 years typically require 1,400 calories.
- Females aged 9 to 13 years typically require 1,600 calories.
- Males aged 9 to 13 years typically require 1,800 calories.
- Females aged 14 to 18 years typically require 1,800 calories.
- Males aged 14 to 18 years typically require 2,200 calories.

General guidance from most governments is that people should ensure they have at least three days' supply of food and water per person stored for a hibernation situation. However, for pandemic situations the food stores should typically be 20 days or more, and it is advisable to resource up to 60 days for a protracted hibernation situation. Under some situations you will be able to replenish some stores, understanding that when leaving your home you expose yourself to potential risks associated with the crisis situation and so should – ideally – limit movement as much as possible.

It is good practice to start rationing early to get you and your family into the habit of conserving food and other scarce resources. This will extend the duration where you can stay healthy while hibernating, while also limiting your need to leave the safety and security of your home.



Ideal Foods for Hibernation and Isolation

Ready-to-eat canned, freeze dried or dry foodstuffs offer the best long-term solution during a hibernation situation, especially if power supplies might be interrupted, making refrigeration unreliable. Some common items you may want to stock-up on include:

- Meat products – ideally canned, frozen, dried or sealed.
- Fruit, vegetables, pulses and beans.
- Rice and pasta products.
- Snack-style bars including cereal, granola, protein and fruit bars.
- Peanut butter.
- Milk powder.
- Baby formula and food.

You should also include comfort or stress foods which support morale during periods of duress.

There are options to preserve perishables, including:

- Parboiling vegetables which are then frozen once cooled;
- Cooking raw meat, poultry, fish and eggs and then freezing, noting that foodstuff must be eaten within a safe timeframe;
- Filling up your freezer with perishables including meat, fish, vegetables, cooked rice, beans and pulses so you have these frozen products to fall back on once all fridge items have been depleted; and
- If freezers fail you can ice pack food to extend the lifespan of perishables, placing the container in a cool dark place.

WARNING: Do not eat food from cans that are swollen, dented or corroded, even though the product may look safe to eat. Do not eat any food that looks or smells abnormal, even if the can looks normal. Frozen items must be properly defrosted and cooked thoroughly before eating.



Using Perishables First

You should discard any perishable food such as meat, poultry, fish, eggs or leftovers that have been above 40° Fahrenheit (or 4°Celsius) for two hours or more. Without electricity or a cold source, food stored in refrigerators and freezers can become unsafe as bacteria in food will grow rapidly at temperatures between 40 and 140 °Fahrenheit. If these foods are consumed then you can become sick, exacerbating your personal crisis situation. Thawed food usually can be eaten if it is still refrigerator-cold, and it can be re-frozen if it still contains ice crystals.



Some tips for keeping food fresh to eat:

- Keep food in covered containers or sealed packaging.
- Discard any food that has been contaminated by flood water.
- Discard any food that has been at room temperature for two hours or more.
- Discard any food that has an unusual odor, color or texture.

To be safe, remember... when in doubt then throw it out!

Refrigeration of Food Resources

When storing food some considerations include:

- Keeping the refrigerator or freezer doors closed for as long as possible.
- Refrigeration will keep food cold during a power loss for about 4 hours IF the door is unopened.
- You can also seek to create your own refrigeration solution by:
 - Going underground. Our ancestors used caves and cellars. If necessary, you can dig deep, or find natural caves already in place where you can store sealed food stores.
 - Using fresh water from a cool stream, especially if it is fresh run-off from melting snow. You can pack food into waterproof containers and leave them in the water to stay cool.

- Using evaporation to cool water and food. You can wrap fabric around a shelving unit and wet it down; placing food on the shelves. You should continue to dampen the fabric to ensure it remains moist. This method works best if the storage is further cooled by a breeze as circulating air increases the amount of evaporation. This type of cooling needs to be in a relatively dry climate.

Using the Zeer pot method which is an evaporative cooler consisting of two unglazed clay pots, with one larger than the other. The small pot is placed inside the larger one and the space between the two pots is filled with sand. Food is placed inside the inner pot and the sand is filled with water. Since the pots are unglazed, the water will soak through them, making the clay wet. Water on the surface of the outer pot will then evaporate, cooling the pots and their contents. By covering the top with a wet cloth then this will increase the overall cooling, helping to keep the food inside fresh.



Water Purification and Drinks



Humans require drinking water on a frequent basis to remain healthy. Dirty or contaminated water can make you sick, and this can lead to dehydration. The average person requires 1 gallon or 3 liters of water per day for drinking, and further water for basic sanitation. Where a water source is dirty or contaminated, or where you are unsure whether it is clean, then you can:

- Use water purification tablets or drops. These destroy harmful bacteria found in contaminated water and protect against illnesses caused by water-borne diseases.
- Boil water using a campfire, stove or other heat source. Over a high heat you should bring water to the boil in a container until you have rolling bubbles on the surface of the water. You should let the bubbles roll for at least five minutes.
- You can buy specialist products that filter water, such as the 'LifeStraw'.
- You can use household bleach which when diluted to 9-parts water to 1-part bleach can be used as a disinfectant. Or in an emergency, you can use certain types of bleach to treat water by using 16 drops of regular household liquid bleach per 1 gallon of water. Do not use scented, color-safe or bleaches with added cleaners. To ensure you use the right amounts of bleach you should use a medicine dropper to measure out the amount of bleach added to the water. Always stir the solution to ensure an even distribution of the active ingredient. This mixture can also be used for washing fruit and vegetables, for cooking food, or for cleaning your teeth.

Other liquid products include:

- Non-perishable pasteurized milk, including long-life UHT.
- Canned juices.
- Ready-mixed infant formula.

Cooking Equipment



Cooking and eating utensils must always be kept clean to prevent sickness. Alternative cooking sources can be used in times of emergency, including candle warmers, chafing dishes, fondue pots or a fireplace. Charcoal grills and camp stoves can be used, but are for outdoor use only. Commercially canned food may be eaten out of the can, without warming, or can easily be warmed in the can by:



- Removing the label.
- Thoroughly washing and disinfecting the can using a diluted solution of 1-part bleach to 9-parts water.
- Opening the can before heating.

Fuels



Various fuels are suitable to aid cooking, these include:

- Open wood fires.
- Bio-fuel systems.
- Gas canisters.
- Charcoal or coal.
- Gel type systems.
- Fuel tablet blocks.



In addition, charcoal and coal dust can also be used to form tablets for burning.

Lighting



Light is important from a safety perspective, as well as enabling work and movement indoors when the power fails, or outside during the hours of darkness. Some considerations include:

- Having a flashlight with extra batteries.
- Using candles, while being mindful of fire hazards.
- Placing matches in a waterproof container.
- Using light sticks - these require no batteries and emit no sparks, are 100% waterproof and do not give off any flames and therefore are not a fire hazard. You should keep to white light where possible.

Warmth



Maintaining warmth will help you sustain your morale, and also reduces the likelihood of hyperthermia and sickness. Keeping dry and having the ability to either find or construct a shelter is your first priority. Other methods of staying warm include:

- Reducing the amount of rooms in use to conserve heating or fuels.
- Hanging blankets or sheets over doorways and windows to reduce the loss of heat.
- Dressing in layers, ensuring you have loose fitting clothes to trap heat in air pockets.
- Keeping your feet warm by using thick socks or layers.
- Keeping your head warm by using hats, scarves or improvised headgear.
- Using foil safety blankets or wool blankets.
- Using old coats, curtains or other fabrics to insulate yourself from the cold.
- Staying close to someone [where social distancing is not an issue] to share body heat.
- Using fires for heat an area, while being mindful of fire hazards.
- Padding the gaps between clothing layers with dry leaves or fabric for insulation.

Health and Hygiene

Health and hygiene are essential during any isolation or hibernation situation, promoting a sense of purpose and positive thinking by maintaining personal standards, as well as a sense of wellbeing. This can be achieved by:

- Using moist towelettes to wipe yourself down and disposing of them in sealed garbage bags.
- Using damp cloths which can then be rinsed out where clean water is available.
- Ensuring waterways (rivers and streams) are kept clean when selecting areas for toilet use so as to avoid contaminating the water source.
- Using fast-flowing water which is cleaner than sluggish or still water.
- Limiting exercise. It is useful to have a good regime of moderate exercise to keep physically and mentally alert, however exercising should be balanced when protecting limited water resources for drinking [due to excessive dehydration], as well as for use for personal hygiene.

Medical and Medicines

Having some first aid knowledge is useful, especially in situations where medical services are overstretched and might be unable to support you. Ideally, you should take a First-Aid course before the crisis occurs. Where this is not possible then accessing training videos or manuals on First Aid can provide some level of knowledge and skills. If others within your family or group have First-Aid skills then you should seek instruction from these individuals. You should ensure you have:

- A First Aid kit with an instructional manual.
- Enough prescription medication required for yourself and your family.
- Over the counter products useful for stomach illnesses to address pain and fevers.
- Documentation related to your medication, including prescriptions and instructions.
- Your eyeglasses or contact lenses.
- The location of local hospitals, clinics, pharmacies and doctor's surgeries.

Morale and Welfare



Being with other people makes any difficult situation better. However, where isolation or social distancing is required [such as during the COVID-19 outbreak] then being able to hear and talk to someone helps to reduce the sense of isolation, and helps to maintain morale. Other methods of maintaining morale include:



- Doing something you enjoy such as reading, playing games, puzzles or engaging in hobbies.

- Creating plays and acting.

- Playing memory games.



- Telling stories, remembering good times, being positive, keeping good thoughts, planning for the future and thinking about “survival!”.

- Maintaining remote communications with colleagues, families and friends while the infrastructure and communication channels are operational.



- Starting a journal and recording your experiences and feelings. Or, writing a story or your memoirs.

- Setting a routine, including creating social time, rest time, sleep time, eating time and work or study time.

- Maintaining a purpose with daily goals, keeping things tidy and setting your self-tasks.

Resource Security



During times of crisis some people can resort to a survival mode which can lead to increased levels of crime. The ‘Social Contract’, that is an agreed set of behaviors within society, can be undermined by a crisis and during extreme situations people may look to steal food, water and other useful items to ensure their own safety and security, or to try and

safeguard the needs of their family. Or, criminals can seek to exploit a weakened law enforcement system for their own personal gain. During periods where instances of crime and social disorder are elevated then you should consider:



- Using purpose-built safes to store valuables where these are available.
- Keeping valuables out of direct lines of sight, such as away from windows and doorways.
- Not talking about what resources you have.
- Hiding critical resources under floorboards, in ceiling areas or in vents.
- Splitting up cash, bank cards and valuables in different places.
- Fitting security lighting and security alarms.
- Setting trip wires with empty cans to alert you to people entering a garden or your home.
- Locking and securing everything as best you can.
- Screwing shut unused access points, while ensuring you still have an escape route.
- Having self-defense items should you be attacked.

Useful Resource Links

The following provides some useful resource links for further guidance:



<https://www.ready.gov/>



<https://www.gov.uk/coronavirus>



<https://www.redcross.org/about-us/news-and-events/news/2020/coronavirus-safety-and-readiness-tips-for-you.html>



<https://www.nhs.uk/conditions/coronavirus-covid-19/>



<https://www.usa.gov/disasters-and-emergencies>



<https://www.redcross.org.uk/get-help/prepare-for-emergencies>